Session objectives

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1. Describe and discuss the pathology and morphology (gross and microscopic) of benign epithelial tumors which include **seborrheic keratoses, acanthosis nigricans, fibroepithelial polyps, and epithelial cysts**

2. Describe and discuss the pathology and morphology (gross and microscopic) of the premalignant epithelial lesion **actinic keratosis**

3. Describe and discuss the pathology and morphology (gross and microscopic) of the malignant epithelial lesions **squamous cell carcinoma and basal cell carcinoma**

4. Describe and discuss the pathology and morphology (gross and microscopic) of melanocytic lesions such as **lentigo and various melanocytic nevi including dysplastic nevi**

5. Describe and discuss the pathology and morphology (gross and microscopic) of **malignant melanoma**

Epithelial tumors

- **Non-melanocytic**
  - **Benign**
    - Seborrheic Keratoses
    - Acanthosis nigricans
    - Fibroepithelial polyp
    - Epithelial cyst
  - **Premalignant**
    - Actinic Keratosis
  - **Malignant**
    - Squamous cell carcinoma
    - Basal cell carcinoma
- **Melanocytic**
  - Solar Lentigo
  - Melanocytic Nevi
  - Dysplastic Nevi
  - Melanoma

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AK: Precursor lesion

- AKs are well-known precursor to squamous cell carcinoma (**SCC**)
- Can develop over **months to years**
  - Can go away on their own or persist for months to years
  - 1-10% of AKs themselves develop into SCCs
- Progression to **full-thickness** cellular/nuclear **atypia** heralds progression into **SCC in-situ**
- Treatment
  - Gentle curettage and/or freezing
  - Topical chemotherapeutics (5-Fluorouracil, Imiquimod)

Squamous Cell Carcinoma (SCC): Overview

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- 2nd most common cancer in the world (>1.5 million cases in US annually)
- **Older** people most affected, **M>W**
- Most important cause is DNA damage from **UV** exposure
  - Roughly proportional incidence to lifetime sun exposure
  - Other risk factors: Industrial carcinogens (tars and oils), chronic ulcers & draining osteomyelitis, old burn scars, arsenical ingestion, ionizing radiation, and (oral cavity) tobacco
- People on **immunosuppression** at an increased risk
- Low rates of invasion and metastases, good prognosis

SCC: Histopathology

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- Pathogenesis
  - Incidence of **p53** dysfunction in **AK** found in Caucasians is high, suggesting this is an early event
  - UV light can damage DNA and cause **immunosuppression** to Langerhans cells
- Clinical findings and histopathology
  - Invasive lesions: Marked hyperkeratosis, **keratin pearls**, **intercellular bridging**, nodules and ulceration

Basal Cell Carcinoma (BCC): Overview

- **Locally aggressive** cutaneous tumor w/ mutations that **activate Hedgehog signaling pathway**
- **Most common invasive cancer** in humans, very **slow growing** (**rarely metastasize**)
- **Most common cancer** in humans (~ 2+ million new cases per year)
- Sun (**UV**) exposed areas most effected
  - Most commonly affect **lightly pigment** patients (blonde hair, blue eyes, pale skin)
  - Risk ↑ w/ immunosuppression
- Clinical appearance
  - **Pearly (waxy) papules**, often with **telangiectasias**
  - **Rolled borders**
  - Central ulceration may occur, less common than SCC
- Histopathology
  - Tumor cells **resemble basal layer** of epidermis
  - Form **nodular** lesions (nests) that **infiltrate** into **dermis**
  - Peripheral **palisading**

BCC: Pathogenesis and treatment

- Pathogenesis
  - Associated with **mutations that activate Hedgehog pathway**; loss of function of **PTCH1 (receptor for sonic hedgehog, SHH)**
    - Loss of function mutation in PTCH1 → Inability to repress transmembrane protein called SMO → ↑ transcription factor, GLI1 → tumor growth
  - **Gorlin** Syndrome: PTCH1 mutation → ↑incidence of BCCs
- Treatment
  - Surgical Excision
    - Wide Local Excision
    - Mohs Micrographic Surgery
    - Cryotherapy
    - Topicals – Imiquimod, 5-Fluorouracil

Solar lentigo: Overview

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- Common **benign localized hyperplasia** of melanocytes
  - Not precancerous, generally do not progress to malignancy
- Exclusively in **sun** exposed areas
- Typically age >40
  - Prevalence ↑ in lightly pigmented individuals
- Typically small (5–10 mm), oval-shaped, tan to brown macules
- Can occur on **skin or mucosa** membranes
- **Lentiginous growth pattern** seen in a subtype of melanomas (lentigo maligna melanoma)
  - Melanocytes proliferate linearly along dermoepidermal junction
    - Rather than forming clusters or nests

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Melanocytic nevus (moles)

- Any neoplasm of melanocytes
- Can be congenital or acquired
- Wide ranging appearance and histology that can change over time
- Typically smaller than malignant melanomas (**<6 mm diameter**)
  - **Symmetric**, **uniform** in color, with **smooth borders**
- ~70% have either **BRAF** or **NRAS mutation**

Saavedra et. al, Fitzpatrick's Clinical Dermatology, 9th edition, 2023

Representative variant forms of melanocytic nevi

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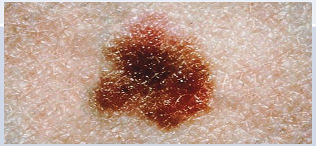
Name		Description
Congenital		**Present at birth**. Can have tuft of dark hair emerging. Can be flat (smaller) or raised (larger). Melanoma can rarely develop within a congenital nevus.
Blue nevus		Acquired, firm dark-blue to grey-black nodule with sharp borders. Localized proliferation of **dermal melanocytes**.
Spitz nevus		Common in **children**. May **appear similar to hemangioma**. Small, dome-shaped, hairless. Often noted for recent, rapid growth.
Halo nevus		Marked by ”**halo**” of depigmentation (leukoderma), surrounding a nevus. **Autoimmune** response against surrounding melanocytes at the dermal-epidermal junction. Can occur in **vitiligo** and indicate incipient vitiligo.
Dysplastic nevus		**Atypical nevi, potential indicator of future melanoma**. Represented by cytological **atypia**, **intraepidermal nests**.

Dysplastic nevus (atypical mole): Overview

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- **Potential precursors** to melanoma
  - Most never progress
  - ↑ risk for melanoma w/ ↑ dysplastic nevi
- Disordered proliferation of **atypical** melanocytes
- Most occur on **exposed skin** (but also covered skin of lightly pigmented individuals)
- Often carry activating **BRAF** or **NRAS** mutations (proto-oncogenes) and loss of function mutations in **CDKN2A** (tumor-suppressor)
- Can be autosomal dominant familial **multiple dysplastic nevi and melanoma syndrome**
  - Hundreds of dysplastic nevi

Melanoma major subtypes

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- **Superficial spreading melanoma**
  - Most common
  - **Best** prognosis, long **radial growth** phase
- **Lentigo maligna melanoma**
  - Higher rate in older individuals
  - **Good** prognosis, **radial growth** phase for years
- **Nodular melanoma**
  - Rapid radial growth & **early vertical** growth
  - **Poorer** prognosis, especially if metastasizes
- **Acral lentiginous melanoma**
  - Most common in **non-white patients**
  - Palms, soles, subungual sites most effected
  - Early vertical growth
  - **Poor** prognosis due to **delayed recognition**

Melanoma diagnosis & prognostic factors

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- **Diagnosis**
  - Excisional **biopsy** to confirm diagnosis & determine thickness (**Breslow depth**) and **ulceration**
  - **Sentinel lymph node** biopsy often done for larger tumors to check for local spread
  - IHC staining & genetic testing
- **Prognosis**
  - **Breslow Depth** (measure from epidermal granular layer top surface to point of maximum tumor thickness)
  - Local & distant **metastases**
  - Mitotic rate
  - Ulceration
  - Tumor infiltrating lymphocytes

TNM cancer staging system for melanoma

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- **Breslow depth** (thickness) and **ulceration**
  - T1: ≤ 1 mm
  - T2: 1.01 to 2 mm
  - T3: 2.01 to 4 mm
  - T4: > 4 mm (Subcategories: a (no ulceration), b (with ulceration): Ex. 4a – without ulceration, 4b – with ulceration)
- Staging
  - Stage 0 – **in-situ** (confined to epidermis)
  - Stage I-II: **Localized or locally advanced** (thicker - ↑ risk than Stage I) primary tumor, **no nodal spread or metastases** (**T1-T4**, N0, M0) ~ 98% 5-year survival
  - Stage III: Regional lymph **node** involvement (T1-T4, **N1-N3**, M0) ~35% 5 year survival
  - Stage IV: Distant **metastases** (Any T, Any N, **M1**)

American Joint Committee on Cancer (AJCC)

Melanoma treatments

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- **Excision** of lesion and surrounding normal tissue (wide local excision)
  - **Margins** of tissue evaluated for tumor (desire clear/clean/negative margins)
    - Indicates surgeon likely removed all tumor, reducing risk of cancer recurrence there
  - Often **sentinel lymph nodes** evaluation guides tumor staging
- **Checkpoint inhibitors**: Blockade of inhibitory receptors, CTLA-4 and PD-1, **enhances T-cell-mediated antitumor** immune responses, leading to improved survival
    - CTLA-4 Inhibitors: **Ipilimumab**
    - PD-1 Inhibitors: **Nivolumab** and **Pembrolizumab**
- Targeted therapies in cancers with known genetic mutations
  - BRAF Inhibitors: In patients with BRAF V600E mutations that drives melanoma growth